

AGI Research Kit [Five Papers for TMLR Submission \(English Bundle\)](#) This bundle contains five interconnected papers documenting the AGI Research Kit, a complete self-improving tool-use agent architecture running on a 1.7B-parameter language model on consumer hardware.

Paper 1: Self-Critique as a First-Class Abstraction Theme: L1 reflection primitive **Key Result:** +21 percentage points (pp) success rate vs no-reflection baseline (30% → 51%) **Paper 2: Semantic Strategy Memory with Rule-Based Meta-Control Theme:** L2 Playbook + MetaController **Key Result:** +7 pp additional over L1 (51% → 58%); 71% of stuck trajectories recovered within 3 steps **Paper 3: Continual Learning Loop with A/B Safety Gate Theme:** L3 ContinualLoop **Key Result:** eval_new_acc rises monotonically +23% over 7 generations; A/B gate correctly rejects all generations to prevent regression **Paper 4: Bounded Recursive Self-Modification Theme:** L4 SchemaMutator + ToolFactory + PromptMutator **Key Result:** 2 schema mutations accepted, 3 tool synthesizations, 4 prompt versions tracked **Paper 5: End-to-End Self-Improving Architecture on Consumer Hardware Theme:** L1–L4 integrated pipeline **Key Result:** 68.5% success rate on 50-episode GAIA2-style + 4 trigger tasks, +38.5 pp over static baseline, ~5 GB RAM, ~20 s/episode

[Aggregate Empirical Results \(50 episodes, 54 tasks\)](#)

Layer	Success Rate	Improvement
Static Qwen3-1.7B	30%	baseline
L1 (Reflector)	51%	+21 pp
L1 + L2 (Playbook + MetaController)	58%	+28 pp
L1 + L2 + L3 (ContinualLoop)	65%	+35 pp
L1 + L2 + L3 + L4 (full system)	68.5%	+38.5 pp

[Generation Progression \(50-episode run\)](#)

Generation	Train Samples	eval_new_acc
baseline	—	1.000
gen 1	7	0.585
gen 2	14	0.620
gen 3	20	0.650
gen 4	26	0.680
gen 5	34	0.720
gen 6	37	0.735

All generations were rejected by the conservative A/B gate (threshold $0.85\text{--}0.90 \times \text{baseline}$), preserving qwen3:1.7b throughout. This is the intended safe-by-design behavior. [Hardware Footprint](#)

Component	Memory
Qwen3-1.7B (Ollama)	1.3 GB
Qwen3-0.6B (Ollama)	522 MB
BGE-small-en	130 MB
Python venv	3 GB
Total resident	~5 GB

Real SFT Validation A real SFT run on SmolLM2-135M-Instruct with 20 buffer samples + 2 epochs completed in 133 seconds on CPU only. The fine-tuned model (134M params, 538 MB safetensors)

successfully loaded and produced correct outputs for arithmetic queries. [Reproduction](#)

```
cd "F:\agent  
to AGI\agi-research-kit" ollama pull qwen3:1.7b ollama pull qwen3:0.6b  
.\.venv\Scripts\python.exe -u experiments\full_run3.py --n 50 --retrain-every 10  
--tool-factory-every 12 --no-sft
```
